Mother-Infant Separation in Squirrel Monkeys Living in a Group

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Two female and 3 male infant squirrel monkeys were separated from their mothers at a mean age of 169 days. All had been group reared. The experiment entailed three 5 day periods: (1) preseparation; (2) separation; and (3) reunion. During the experiment, each infant was observed for 10 min per day. Frequency and duration measures were recorded for activity play, locomotion, object manipulation, and disturbance. Social behaviors recorded were contact play, mixed play, competition play, sex play, affiliation, display, and aggression.

Changes noted during separation were depression of activity play, contact play, mixed play, object manipulation, and sex play. Affiliative behaviors and locomotion increased during separation.

After reunion, locomotion and affiliation returned to baseline levels while contact play, mixed play, and object manipulation remained depressed.

Systematic investigations of the effects of mother-infant separation in nonhuman primates began only 10 years ago. All of the experiments conducted thus far, except one (Kaplan, 1970), have employed Old World monkeys, specifically, rhesus, pigtail, and bonnet macaques (*Macaca mulatta, M. nemestrina*, and *M. radiata*, respectively); Kaplan used squirrel monkeys, (*Saimiri sciureus*). Despite the variety of separation techniques used, the results of the separation experiments with rhesus monkeys are remarkably consistent. Studies performed by Seay, Hansen, and Harlow (1962), Seay and Harlow (1965), Spencer-Booth and Hinde (1967, 1971a, 1971b), and Hinde and Spencer-Booth (1971) revealed a cluster of behavioral changes shown by rhesus monkey infants following separation from, and subsequent reunion with, their mothers. During periods of separation from their mothers, infant rhesus monkeys increased distress calling and crying and decreased locomotion and social play. Reunion with their mothers produced initial increases in close maternal contact compared to preseparation baseline levels. Recovery of social play and locomotion usually occurred within a few days after reunion.

Comparative studies of maternal separation effects on group living pigtail and bonnet monkey infants (Kaufman & Rosenblum, 1967, 1969) revealed various interspecies differences. For example, when their mothers were removed from the group, pigtail monkeys demonstrated depression of play and locomotor activity,

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similar to rhesus monkeys, whereas the bonnet monkey infants' behavior remained unchanged.

In Kaplan's (1970) experiment, individually housed squirrel monkey mother-infant dyads were separated for 7 days and then reunited. During separation, the infants vocalized more than before separation, but their locomotion levels remained unchanged. Following reunion with their mothers, the infants' vocalizations returned to preseparation levels and locomotion decreased as a result of the infants spending more time on their mothers than prior to separation.

The experiments mentioned above suggest that the effects of maternal separation on infants' behavior differ somewhat across primate species. As Rosenblum & Kaufman (1968) indicated, separation studies are valuable for comparing the relative influences of the mother and the group upon the social development of young primates. Infant pigtail and rhesus monkeys receive early social influence almost exclusively from their mothers, whereas, bonnet monkey infants receive a large amount of early social experience from other adult and juvenile members of their primary group, the group with which they are reared (Rosenblum & Kaufman, 1967).

Kaplan's (1970) subjects were squirrel monkeys whose early social experiences came solely from their mothers. His separation procedure also involved social isolation in addition to maternal separation; the infants were housed singly during separation. The present experiment was designed to investigate maternal separation in squirrel monkeys reared in a group.

Method

The subjects for this experiment were 3 male and 2 female squirrel monkey infants (*S. sciureus*) group reared from birth with their mothers in a communal cage within the laboratory. These infants and their mothers were the only members of the group. The subjects were not handled or otherwise disturbed until separation from their mothers. The mean age of the subjects at separation was 169 days (range, 153-186 days).

The cage in which the data were collected and in which the subjects were reared from birth measured 2m on each edge. Three shelves attached step-wise to the back of the cage, 2 perches, and 1 swing furnished the cage. The room in which the cage was located was separated from the main laboratory squirrel monkey colony. Plastic toys (e.g., small airplanes, balls, and blocks) were provided for the subjects.

The data recording apparatus consisted of key-operated frequency and duration counters and 4 foot-operated electric timers. These devices were located on an elevated platform placed 1m in front of the cage.

The subjects and their mothers were fed twice daily, once at 0700 hours by the laboratory staff, and again at 1600 hours by the experimenters. Ambient temperature was maintained at approximately 31°C, and relative humidity at 55-60%. The normal 12-hr light-dark cycle with lights on from 0700-1900 hours were maintained throughout the experiment.

The experiment consisted of 3 phases:

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Preseparation. Data were collected for 5 days prior to the mothers' removal from the cage. During this period, the only change in the subjects' daily routine was the experimenters' presence during data collection.

Separation. On the day of separation, the subjects' mothers were fed sugar cubes containing 4 mg of phencyclidine hydrochloride, (Sernylan, Phillips-Roxane, Inc.), an immobilizer, in order to reduce the effects of the separation procedure on the mothers, infants, and experimenter. This endeavor was only partially successful since only 2 of the 5 mothers ingested enough of the drugged sugar cube to produce the desired reduced responsiveness. Regardless of whether their mothers had been drugged or not, all infants were handled during the separation procedure. (The infants of the 2 drugged mothers were lifted off their mothers.) Infants not accustomed to being handled emit a high-pitched peeping vocalization, which in turn elicits shrieking and general arousal from other squirrel monkeys in the vicinity. Although separation of the dyads in which the mothers were not drugged resulted in more resistance from these mothers, all infants were exposed to the commotion as well as being handled. Differences in behavior between the infants whose mothers had been drugged vs. those whose mothers had not been drugged were not apparent either qualitatively or quantitatively at any point of the separation or reunion phases. Separation was performed at 1000 hours on the 6th day of the experiment, and was carried out by removing all mothers from the group cage simultaneously and placing them in another group cage in a separate room; the infants remained in a group. Data were collected on the day of separation in the same manner as during preseparation. The infants were separated from their mothers for 5 days.

Reunion. At 1000 hours on the day of reunion (Day 11 of the experiment), all mothers were simultaneously replaced in the group cage containing their infants. Observations on this day and on the remaining days were carried out as they had been previously.

Observation periods always began promptly at 1500 hours and continued for 50 min. Each subject was observed on each day of the experiment during a 10 min scoring period with the order of observation of subjects randomized across days. One experimenter (B.C.J.) was the sole data recorder, making interobserver reliability assessment unnecessary.

Durations of infant nonsocial and infant-infant social behaviors were recorded during all phases of this experiment.

Infant nonsocial behaviors measured were:

Activity play:	vigorous locomotion, swinging or jumping, or any combination of these.
Locomotion:	movement of a distance greater than one body length.
Object direct:	visual (within approximately 7 cm) or tactual exploration, or play directed toward any part of the cage or objects placed within.

Infant-infant social behaviors measured were:

Contact play:	initiation or reception of wrestling and/or play fighting.
Mixed play:	initiation or reception of play involving predominately chasing.
Competition:	initiation or reception of an attempt to take away toys or other objects from or by another infant.
Sex play:	initiation or reception of mounting, with or without thrusting.
Affiliation:	initiation or reception of approach or huddling.

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Display: initiation (at any distance from another infant), or reception (from within 15 cm), of hind limb abduction, with or without vocalization.

Aggression: biting by subject or by another infant so as to elicit avoidance, distress vocalization, or retaliatory action.

Disturbance: crying or shrieking for any reason, observed or not.

During preseparation and reunion, separate durations of the time spent by the infants on their mothers, off their mothers and at the same level of the cage, and off their mothers and at another level of the cage were recorded. Level was defined as vertical separation greater than 30 cm.

Results

Infant nonsocial and infant-infant social behaviors were evaluated by analyses of variance for a 2-factor experiment with sex as the between-subjects variable and separation as the within-subjects variable. Subsequent comparisons were made using Scheffé's (1953) Multiple Comparisons Method, and as suggested by Scheffé, results were considered reliable if $p < .10$. "Days" was not included in the ANOVA as an additional within-subjects variable because of recording equipment failure on 2 days during the reunion phase. Instead, in order to assess changes within the separation and reunion periods for those behaviors which showed reliable changes for the periods as a whole by ANOVA's, the last days of separation and reunion were compared with each other, and with the last day of preseparation. These comparisons were made using the Wilcoxon Signed Ranks Test (Wilcoxon & Wilcox, 1964). Durations of the time spent by the infants on mother, off mother at the same level of the cage, and off mother at another level of the cage before separation and after reunion were also analyzed by means of the Wilcoxon Signed Ranks Test.

High intersubject variability was noted for many of the behavioral categories in this experiment. This variability presents 2 problems for the application of the analysis of variance as a statistical tool. First, heterogeneity of covariance may occur in the separation data, the repeated measure. Heterogeneity of covariance is a positively biasing factor, viz., may lead to Type I errors. This can be partially overcome by the use of summary measures in the ANOVA (Hays, 1963). The basic data for the ANOVA's were mean scores in the present case. Second, if heterogeneity of covariance were slight, a large amount of intersubject variability would decrease the power of the ANOVA's. When as few degrees of freedom as were available in this experiment are considered, the use of the analysis of variance represents a conservative approach.

Figure 1 illustrates changes in the nonsocial behaviors scored during separation and after reunion. During the separation period, activity play decreased ($p < .10$). Following reunion, activity increased nonsignificantly compared to the separation period levels. Object manipulation also decreased during separation ($p < .10$) and remained depressed following maternal reunion ($p < .10$). Locomotion increased during separation ($p < .01$); after reunion it returned to baseline levels ($p < .01$).

Changes in infant-infant social behaviors during separation and after reunion are illustrated in Figure 2. During separation, reliable decreases were observed in contact play ($p < .01$), mixed play ($p < .05$), and sex play ($p < .10$). Following maternal

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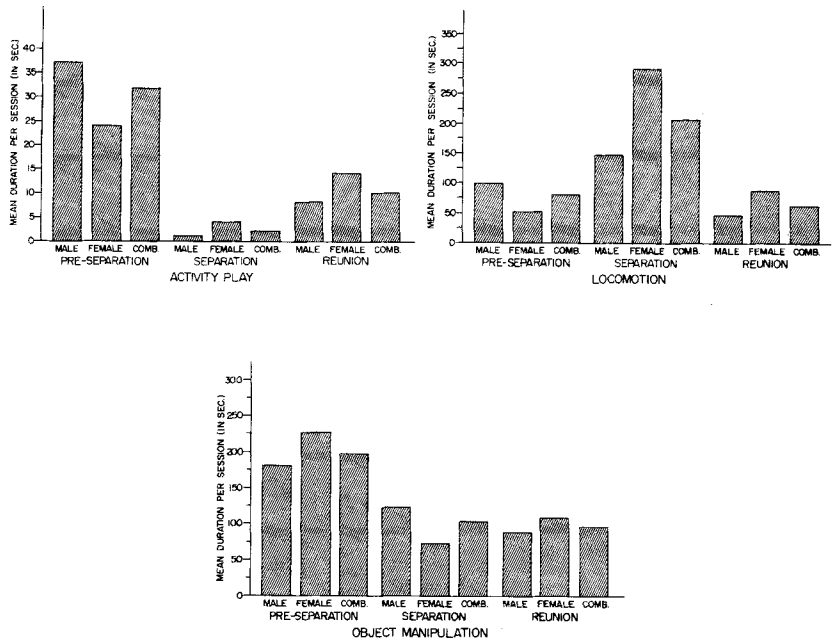


Fig. 1. Mean duration per observation session of nonsocial behaviors scored.

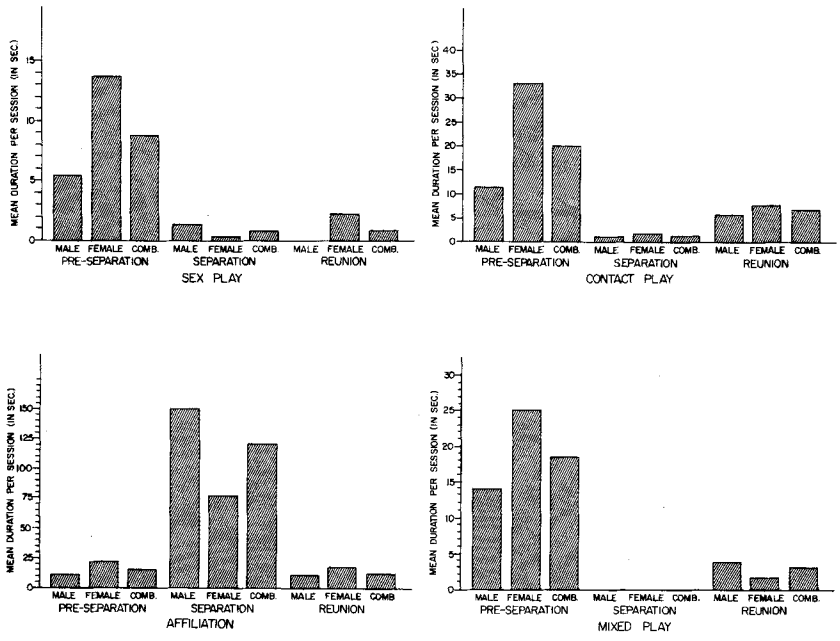


Fig. 2. Mean duration per observation session of social behaviors scored.

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TABLE 1. Mean Duration (sec) per Observation Session of Scored Behaviors.

	Preseparation			Separation			Reunion			Post Hoc Comparisons, Combined Means		
	M	F	Comb.	M	F	Comb.	M	F	Comb.	P-S	S-R	P-R
Activity Play	37.33	23.75	31.90	1.00	3.50	2.00	8.07	14.05	10.46	.10 ↓	N.S.	N.S.
Locomotion	99.00	52.20	80.30	148.80	292.92	206.46	47.50	84.90	62.48	.01 ↑	.01	N.S.
Contact Play	11.27	33.30	20.08	.94	1.20	1.04	5.90	7.70	6.62	.01 ↓	.01	.01
Mixed Play	13.77	25.05	18.28	.0	.0	.0	3.80	1.60	2.92	.05 ↓	.10	.05
Competitive Play	11.67	20.04	15.16	21.20	7.20	15.60	7.90	10.20	8.82	N.S.	N.S.	N.S.
Sex Play	5.34	13.80	8.72	1.30	.30	.82	.0	2.30	.92	.10 ↓	N.S.	N.S.
Affiliate	11.14	21.60	15.32	150.87	76.10	120.96	11.80	16.75	13.78	.05 ↑	.05	N.S.
Display	1.53	.0	.92	.30	.0	.18	2.43	2.25	2.36	N.S.	N.S.	N.S.
Aggression	.33	2.40	1.16	4.10	3.95	3.96	.90	.35	.50	N.S.	.10	N.S.
Disturbance	3.57	.0	2.14	14.07	53.00	29.64	1.60	3.40	2.32	N.S.	N.S.	N.S.
Object Manipulation	178.50	224.85	197.04	123.79	72.9	103.43	88.40	107.30	95.98	.10 ↓	N.S.	.10

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reunion, contact play and mixed play increased over the levels observed during separation ($p < .01$, $p < .10$, respectively), but were still depressed in comparison to the preseparation levels ($p < .01$, $p < .05$, respectively). The category of contact play revealed a reliable interaction between the factors of Sex and Separation ($p < .025$). Females showed higher levels of contact play than males during preseparation ($p < .05$), but no separation or reunion period differences were observed between the sexes. Infant-infant affiliative behaviors, i.e., approach and huddling, increased significantly during separation ($p < .05$), then returned to the preseparation baseline levels following reunion ($p < .05$).

Competition play, display, aggression, and disturbance (crying, shrieking) did not change reliably between the experimental periods. (See Table 1 for mean scores for nonsocial and social behaviors.)

A comparison of the last day of preseparation with the 1st day of the separation period for duration of disturbance (Table 2) revealed a reliable increase in this category ($p < .05$; Wilcoxon & Wilcox, 1964). However, this behavior decreased significantly between the 1st and last days of separation ($p < .05$), indicating recovery of disturbance during the separation period.

Duration scores recorded on the last day of preseparation, separation, and reunion for activity play, locomotion, contact play, mixed play, sex play, affiliation, and object manipulation are presented in Table 3. Application of the Wilcoxon Signed-Ranks Test to these data indicated reliably persistent depression effects throughout separation in activity play, contact play, and sex play ($p < .05$). Locomotion scores, as previously indicated in Figure 1, increased for the group as a whole during separation. The large variation of scores on the last day of separation was the result of an observed increase in locomotion across days for the 2 females, Subjects 1 and 3, and a general decrease in locomotion for the 3 male subjects. Even though all the subjects showed an increase in affiliation scores during the separation period as a whole, no consistent changes within the separation period were discovered. This pattern was also generally true for the category of object manipulation, an exception being Subject 5, whose object manipulation scores increased during separation, even surpassing his highest preseparation scores in this category.

Infant nonsocial and infant-infant social behaviors observed during reunion should be considered in the context of the responses of the infants to their mothers immediately following reunion. The difference between the infants' behavior toward their mothers immediately after reunion, and the last day of preseparation was a

TABLE 2. Mean Duration of Crying (sec) for each Subject for Selected Days of the Experiment.

Subject	Preseparation	Separation		Reunion	
	Last Day	First Day	Last Day	First Day	Last Day
1	0	0	2	64	0
2	0	211	0	0	0
3	0	412	45	0	0
5	0	47	0	0	0
6	9	114	6	0	0

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TABLE 3. Duration (sec) of Selected Behaviors on the Last Day of Preseparation, Separation, and Reunion.

Subject	Activity Play			Locomotion			Contact Play			Mixed Play			Sex Play			Affiliation			Object Manipulation		
	P	S	R	P	S	R	P	S	R	P	S	R	P	S	R	P	S	R	P	S	R
1	11.0	.0	9.5	43.0	514.0	125.0	80.0	.0	19.5	79.5	.0	1.5	.0	.0	.0	6.5	18.5	4.5	209.5	42.5	148.5
2	88.0	.0	.0	87.0	8.0	65.0	3.0	.0	.0	4.0	.0	.0	4.5	.0	.0	4.5	35.0	2.0	293.5	.0	121.5
3	23.5	26.0	13.0	90.0	533.0	99.5	59.0	.0	8.0	52.4	.0	.0	1.5	.0	.0	28.0	4.0	13.5	183.5	6.0	40.0
4	4.0	.0	4.0	59.0	55.5	69.0	16.5	.0	4.0	.0	.0	2.0	20.0	7.5	.0	44.5	39.0	18.5	249.0	331.5	119.0
5	87.5	1.0	5.0	79.5	103.0	77.5	60.5	1.0	16.0	42.0	.0	15.0	16.0	.0	.0	8.0	183.0	7.5	77.5	87.0	234.0

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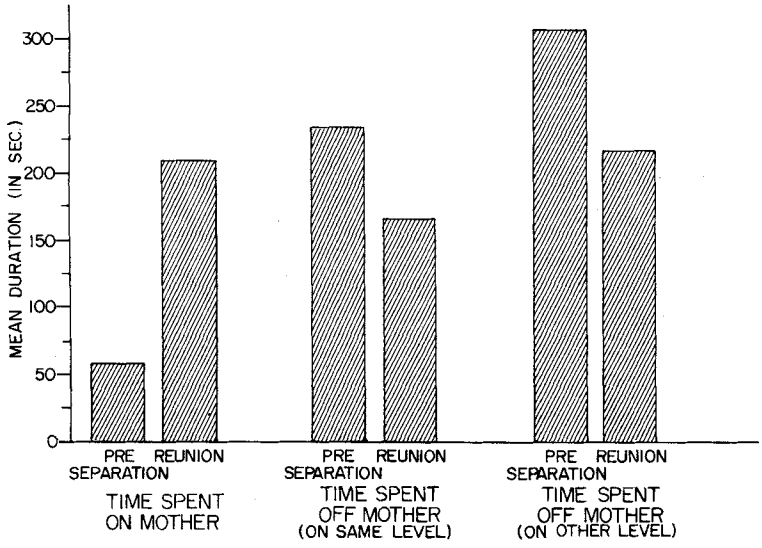


Fig. 3. Mean duration per observation session of the time spent by infants on their mothers, off their mothers at the same level of the cage, and off their mothers at other levels of the cage.

striking increase in maternal contact. Figure 3 illustrates comparisons of mean durations per observation session of the location of the infants, relative to their mothers, during preseparation and reunion.

This increase in time spent on the mothers ($p < .05$; Wilcoxon & Wilcox, 1964), and the concomitant decreases in time spent off the mothers at the same level (n.s.) and other levels of the cage ($p < .05$), were largely accounted for by the infants' behavior on the day of reunion and the day following. On these 2 days, all the infants spent at least half of the observation session asleep on their mothers. In contrast, on the last day of preseparation, none of the infants was seen on its mother. During the last 3 observation periods following reunion, all of the infants began to leave their mothers for longer durations, but during the final observation session following reunion, none of the infants was observed clinging to its mother.

The relatively large amount of time spent by the infants on their mothers during the first 2 days after reunion was reflected in the generally lower mean durations noted for all other behaviors during reunion as compared to preseparation levels (Table 1). The previously mentioned recording equipment failure occurred on Days 3 and 4 following reunion and prevented recorded observations on those 2 days. On Days 5, 6, and 7 following reunion, all the infants began to interact once again with environmental stimuli, including each other. However, contact play, mixed play, sex play, and, perhaps to a lesser extent, activity play were still depressed on the 7th day after reunion compared to the day prior to maternal separation ($p < .05$).

Discussion

A frequently cited descriptive model for maternal separation studies is that of Bowlby (1961), according to which separation of an infant from its mother activates 3

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stages of organized and innate patterns of behavior, descriptively labelled "protest," "despair," and "detachment." Protest was defined as an increase in crying and activity immediately following separation. Despair follows protest and was defined as general behavioral depression. Detachment occurred when the infant finally accepted the permanence of the loss of its mother and was manifested in the delay and/or resistance with which some human infants re-established maternal contact.

The behaviors emitted by the infants immediately after separation in the present experiment certainly defined protest. The recorded increases in crying, which diminished throughout the separation period, and increases in locomotion, which in most subjects appeared as stereotyped pacing, were consistent with the results obtained with other species reared and tested in a group. Kaplan's (1970) infant squirrel monkeys evinced no such increases in activity, defined by him as a cluster of behaviors including "... any movement involving locomotion." Kaplan's subjects spent large amounts of time engaging in activity, relative to the amount of time he spent observing them both prior to and after separation. Either because the spatially and socially restrictive environment employed by Kaplan prevented the infants' diversification of active behaviors, or Kaplan's activity measure was too broad, activity, as a dependent measure, appeared to have been insensitive to his separation treatment.

With respect to Bowlby's despair behaviors (1961), the virtual disappearance of social play and reduction of activity play and object manipulation following separation in the current study correspond with the depressive separation syndrome reported in rhesus and pigtail macaques. However, the extreme withdrawal, viz., solitary crouching and huddling, reported by other investigators was never observed in any of the subjects in the present experiment. Inactivity, such as sleeping, occurred rarely and was observed only in huddles of 2-5 infants.

In the present investigation, as in most other nonhuman primate separation studies no detachment occurred; the infants returned to their mothers immediately upon reunion.

The separation procedure incorporated in this experiment differed in some respects from procedures employed by other investigators studying intact groups. In our case all the mothers were simultaneously removed from an established group of mother-infant dyads. In the previous studies, separation of group-living dyads involved removal of only one mother at a time from the group, thereby leaving the observed infant with a group of peers and adults. The demonstration of protest and severe, persistent depression of complex social behaviors in the present experiment were congruent with the results obtained from pigtail and rhesus macaque separation studies, despite this procedural difference. We thus conclude that squirrel monkey infants can be induced to demonstrate many of the clinical signs of primate anaclitic depression, and therefore are likely to be suitable subjects for future experimental depression research.

Notes

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